

## ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018

First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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## 1. GATE DA 2025

### A. Formulae

#### 1.1.1 Indicator Summation for Thresholding

$$N = \sum_{i=1}^M \sum_{j=1}^N \mathbb{I}(U_{ij}) \quad (1.1.1.1)$$

**Solution:** The total count of elements in a matrix  $\mathbf{U}$  satisfying a threshold  $c$  is obtained by summing the indicator function  $\mathbb{I}$ . The values of  $\mathbb{I}$  operate as a step function defined as:

$$\mathbb{I}(U_{ij}) = \begin{cases} 1, & U_{ij} \leq c \\ 0, & U_{ij} > c \end{cases} \quad (1.1.1.2)$$

#### 1.1.2 Geometric Scaling under Dilation

$$A' = (1 + \alpha) A \quad (1.1.2.1)$$

**Solution:** For a rectangle with length  $L$  and width  $W$ , Area  $A = LW$ . If  $W$  is scaled by a factor of  $(1 + \alpha)$ :

$$W' = (1 + \alpha) W \quad (1.1.2.2)$$

Substituting (1.1.2.2) into the area equation:

$$A' = LW' = L[(1 + \alpha) W] = (1 + \alpha) LW = (1 + \alpha) A \quad (1.1.2.3)$$

yielding (1.1.2.1).

#### 1.1.3 Injectivity of Strictly Unimodal Functions

$$x_1 < c < x_2 \quad \text{for} \quad f(x_1) = f(x_2) \quad (1.1.3.1)$$

**Solution:** If  $f(x)$  strictly increases on  $[0, c]$  and strictly decreases on  $[c, \infty)$ , each branch is individually injective (one-to-one). If two distinct points  $x_1$  and  $x_2$  yield the same function value, they cannot lie on the same monotonic branch, meaning the peak  $c$  must lie strictly between them.

#### 1.1.4 Exponent Laws

$$\frac{a^u}{a^v} = a^{u-v}, \quad (a^x)^y = a^{xy} \quad (1.1.4.1)$$

#### 1.1.5 Conditional Expectation of random variables X and Y

$$E[X] = \sum_k k p_X(k) \quad (1.1.5.1)$$

$$E[X | Y = y] = \sum_k k p_{X|Y}(k | y) \quad (1.1.5.2)$$

#### 1.1.6 Computational Complexity of Back-Substitution For an $n \times n$ upper triangular matrix

system  $\mathbf{U}\mathbf{x} = \mathbf{b}$ , variables are obtained sequentially:

$$x_i = \frac{1}{U_{ii}} \left( b_i - \sum_{j=i+1}^n U_{ij}x_j \right) \quad (1.1.6.1)$$

### 1.1.7 Eigenvalues of Constant Row-Sum Matrices and System Solvability

$$\mathbf{A}\mathbf{1} = \mathbf{1} \cdot \mathbf{1}, \quad (\mathbf{A} - \mathbf{I})\mathbf{1} = \mathbf{0} \cdot \mathbf{1} \quad (1.1.7.1)$$

**Solution:** Let  $\mathbf{a}_i^\top$  denote the  $i$ -th row of  $\mathbf{A} \in \mathbb{R}^{n \times n}$ . Because the sum of the elements in each row is 1, multiplying the row by the all-ones column vector  $\mathbf{1} = (1 \ 1 \ \dots \ 1)^\top$  yields:

$$\mathbf{a}_i^\top \mathbf{1} = 1, \quad \forall i \in \{1, 2, \dots, n\} \quad (1.1.7.2)$$

Grouping these row equations together to reconstruct the full matrix  $\mathbf{A}$  gives:

$$\mathbf{A}\mathbf{1} = \begin{pmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \vdots \\ \mathbf{a}_n^\top \end{pmatrix} \mathbf{1} = \begin{pmatrix} \mathbf{a}_1^\top \mathbf{1} \\ \mathbf{a}_2^\top \mathbf{1} \\ \vdots \\ \mathbf{a}_n^\top \mathbf{1} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1} = \mathbf{1} \cdot \mathbf{1} \quad (1.1.7.3)$$

This demonstrates that the vector  $\mathbf{1}$  is an eigenvector of  $\mathbf{A}$  associated with the eigenvalue  $\lambda = 1$ .

To find the eigenvalue of  $\mathbf{A} - \mathbf{I}$  corresponding to the same eigenvector  $\mathbf{1}$ :

$$(\mathbf{A} - \mathbf{I})\mathbf{1} = \mathbf{A}\mathbf{1} - \mathbf{I}\mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0} = \mathbf{0} \cdot \mathbf{1} \quad (1.1.7.4)$$

Thus,  $\mathbf{A} - \mathbf{I}$  possesses an eigenvalue of 0, meaning it is a singular matrix ( $|\mathbf{A} - \mathbf{I}| = 0$ ).

Number of Solutions for Linear Systems:

- For the system  $(\mathbf{A} - \mathbf{I})\mathbf{x} = \mathbf{b}$ : Because  $\mathbf{A} - \mathbf{I}$  is singular ( $\text{rank} < n$ ), the system will have infinitely many solutions if  $\mathbf{b}$  is in the column space of  $\mathbf{A} - \mathbf{I}$  (i.e.,  $\text{rank}(\mathbf{A} - \mathbf{I}) = \text{rank}([\mathbf{A} - \mathbf{I} \mid \mathbf{b}])$ ), or no solution if it is not.
- For the system  $\mathbf{A}\mathbf{x} = \mathbf{b}$ : The number of solutions depends entirely on the remaining unknown eigenvalues of  $\mathbf{A}$ . If all other eigenvalues are non-zero ( $|\mathbf{A}| \neq 0$ ), the system has a unique solution. If 0 happens to also be an eigenvalue of  $\mathbf{A}$ , the matrix is singular, resulting in either no solution or infinitely many solutions

### 1.1.8 Maclaurin Series of Hyperbolic Sine

$$\sinh(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} \quad (1.1.8.1)$$

**Solution:** Expanding  $e^x$  and  $e^{-x}$  via Maclaurin series:

$$f(x) = \frac{1}{2} \left( \sum_{n=0}^{\infty} \frac{x^n}{n!} - \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} \right) \quad (1.1.8.2)$$

$$= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \quad (1.1.8.3)$$

### 1.1.9 Relational Algebra Natural Join

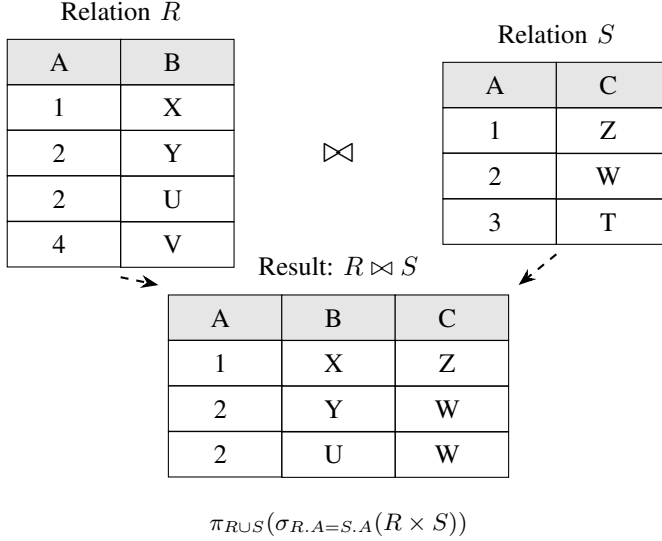


Fig. 1.1.9.1: Pictorial representation of Relational Algebra Natural Join

$$R \bowtie S = \pi_{R \cup S}(\sigma_{R.A=S.A}(R \times S)) \quad (1.1.9.1)$$

**Solution:** The equation describes the mechanical steps of combining two database tables using relational algebra operators:

- $R$  and  $S$ : The two input relations (data tables) being operated on.
- $\bowtie$  (Natural Join): The overarching operator that merges  $R$  and  $S$  by matching rows with identical values in their common columns, outputting a single combined table with no duplicate columns.
- $R \times S$  (Cartesian Product): The foundational operation that pairs every single row in table  $R$  with every single row in table  $S$ , generating all possible combinations.
- $\sigma_{R.A=S.A}$  (Selection): The row-filtering step. The  $\sigma$  operator evaluates the Cartesian product and retains only the rows where the value in the shared attribute  $A$  matches between both tables.
- $\pi_{R \cup S}$  (Projection): The column-filtering step. The  $\pi$  operator dictates which columns appear in the final result. The subscript  $R \cup S$  ensures that all unique

columns from both tables are kept, effectively stripping out the redundant duplicate common column ( $S.A$ ) generated by the Cartesian product.

#### 1.1.10 Linear Probing in Hash Tables

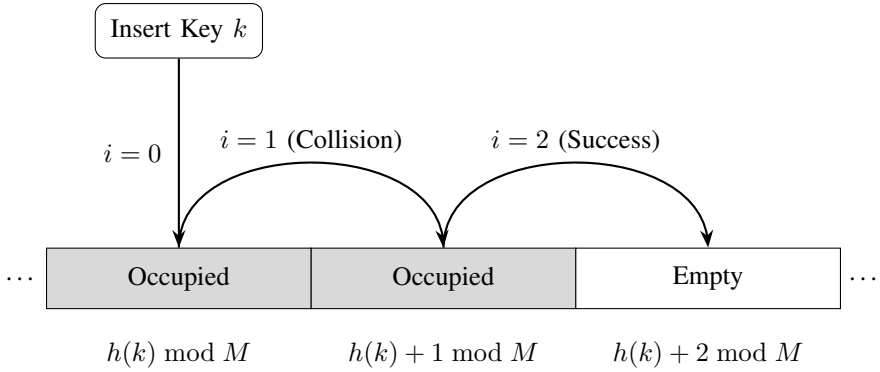


Fig. 1.1.10.1: Pictorial representation of Linear Probing in Hash Tables

$$h(k, i) = (h(k) + i) \pmod{M} \quad (1.1.10.1)$$

The terms in the probing function represent:

- $k$ : The original key or piece of data to be stored.
- $h(k)$ : The primary hash function that calculates the initial target index.
- $i$ : The probe attempt number, starting at 0 and incrementing linearly for each collision encountered.
- $M$ : The total allocated size of the hash table.
- $\pmod{M}$ : The modulo operation ensuring the probe sequence wraps around to index 0 if the end of the table is reached.
- $h(k, i)$ : The final calculated storage index for the  $i$ -th probe attempt.

#### 1.1.11 Median of a Continuous Distribution

Let  $f_X(x)$  be the probability density function. The median  $m$  divides the total probability mass strictly in half:

$$\Pr(X \leq m) = 0.5 \quad (1.1.11.1)$$

$$\int_{-\infty}^m f_X(x) dx = 0.5 \quad (1.1.11.2)$$

Because the integral of  $f_X(x)$  from  $-\infty$  to  $m$  is exactly  $F_X(m)$ , it follows that:

$$F_X(m) = \int_{-\infty}^m f_X(x) dx = 0.5 \quad (1.1.11.3)$$

#### 1.1.12 General Moments of the Standard Normal Distribution

The expected value  $E[Z^n]$  for a standard normal variable  $Z \sim \mathcal{N}(0, 1)$  is defined

by the integral:

$$E[Z^n] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^n e^{-z^2/2} dz \quad (1.1.12.1)$$

For odd powers ( $n = 2k + 1$ ), the integrand is an odd function symmetric about the origin:

$$E[Z^{2k+1}] = 0 \quad (1.1.12.2)$$

For even powers ( $n = 2k$ ), integrating by parts ( $u = z^{n-1}$ ,  $dv = ze^{-z^2/2}dz$ ) yields:

$$\int_{-\infty}^{\infty} z^n e^{-z^2/2} dz = \left[ -z^{n-1} e^{-z^2/2} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} (n-1) z^{n-2} e^{-z^2/2} dz \quad (1.1.12.3)$$

The boundary term decays to zero. Multiplying by the normalization constant establishes the recurrence relation:

$$E[Z^n] = (n-1) E[Z^{n-2}] \quad (1.1.12.4)$$

Expanding recursively to the base case  $E[Z^0] = 1$  gives the expanded product:

$$E[Z^n] = (n-1) \cdot (n-3) \cdots 3 \cdot 1 \quad (1.1.12.5)$$

1.1.13 Let the random variable  $Z$  follow a standard normal distribution.

It follows that

$$E[Z] = 0 \quad (1.1.13.1)$$

$$E[Z^2] = 1 \quad (1.1.13.2)$$

Show that for

$$X = aZ + b \quad (1.1.13.3)$$

It follows that

$$E[X] = aE[Z] + b \quad (1.1.13.4)$$

$$\text{Var}(X) = a^2 \text{Var}(Z) \quad (1.1.13.5)$$

**Solution:** The step-by-step derivations for the expectations are:

$$E[X] = E[aZ + b] \quad (1.1.13.6)$$

$$= aE[Z] + b \quad (1.1.13.7)$$

$$= a(0) + b \quad (\text{using (1.1.13.1)}) \quad (1.1.13.8)$$

$$= b \quad (1.1.13.9)$$

Then,

$$X - E[X] = (aZ + b) - b \quad (1.1.13.10)$$

$$= aZ \quad (1.1.13.11)$$

Similarly,

$$E[(X - E[X])Z] = E[aZ \cdot Z] \quad (1.1.13.12)$$

$$= aE[Z^2] \quad (1.1.13.13)$$

$$= a(1) \quad (\text{using (1.1.13.2)}) \quad (1.1.13.14)$$

$$= a \quad (1.1.13.15)$$

Yielding,

$$E[(X - E[X])^2] = E[(aZ)^2] \quad (1.1.13.16)$$

$$= a^2 E[Z^2] \quad (1.1.13.17)$$

$$= a^2(1) \quad (\text{using (1.1.13.2)}) \quad (1.1.13.18)$$

$$= a^2 \quad (1.1.13.19)$$

#### 1.1.14 Tail Probability of Exponential Distribution

$$\Pr(X \geq x) = \int_x^\infty \lambda e^{-\lambda t} dt = e^{-\lambda x} \quad (1.1.14.1)$$

1.1.15 Let  $A = [a_1, a_2, \dots, a_n]$  and  $B = [b_1, b_2, \dots, b_m]$  represent sequences, and let  $\leftarrow$  denote in-place modification. The python functions map to the following mathematical transformations:

$$A.\text{extend}(B) \implies A \leftarrow [a_1, \dots, a_n, b_1, \dots, b_m] \quad (1.1.15.1)$$

$$A.\text{append}(B) \implies A \leftarrow [a_1, \dots, a_n, B] \quad (1.1.15.2)$$

$$A.\text{update}(B) \implies A \leftarrow A \cup B \quad (\text{where } A \text{ and } B \text{ are sets}) \quad (1.1.15.3)$$

$$A.\text{insert}(i, B) \implies A \leftarrow [a_1, \dots, a_i, B, a_{i+1}, \dots, a_n] \quad (1.1.15.4)$$

#### 1.1.16 Differentiability Rules

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c) \quad (1.1.16.1)$$

If  $f$  and  $g$  are differentiable at  $c$ , their sum, difference, and product are strictly differentiable at  $c$ . The quotient is differentiable if  $g(c) \neq 0$ . Composition  $f \circ g$  requires  $f$  to be differentiable at  $g(c)$ , which is not universally guaranteed.

#### 1.1.17 Properties of Orthonormal Sets

$$\mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad (1.1.17.1)$$

**Solution:** An orthonormal set of vectors in  $\mathbb{R}^n$  is unconditionally linearly independent. Orthonormal bases in  $\mathbb{R}^n$  are not unique because applying any rotation matrix generates a new valid orthonormal basis.

#### 1.1.18 Binary Search Complexity

$$T(n) = T(n/2) + O(1) \quad (1.1.18.1)$$

$$T(n) = O(\log n) \quad (1.1.18.2)$$

**Solution:** Binary search operates by halving the search space at each step. Let  $c$  represent the constant  $O(1)$  time required to access the middle element. Expanding the recurrence relation iteratively yields:

$$T(n) = T\left(\frac{n}{2}\right) + c \quad (1.1.18.3)$$

$$= \left(T\left(\frac{n}{4}\right) + c\right) + c \quad (1.1.18.4)$$

$$= T\left(\frac{n}{2^2}\right) + 2c \quad (1.1.18.5)$$

$$= T\left(\frac{n}{2^k}\right) + kc \quad (1.1.18.6)$$

The recursion strictly terminates when the array size reduces to 1.

$$\frac{n}{2^k} = 1 \quad (1.1.18.7)$$

$$n = 2^k \quad (1.1.18.8)$$

$$k = \log_2 n \quad (1.1.18.9)$$

Substituting  $k$  back into the expanded equation:

$$T(n) = T(1) + c \log_2 n \quad (1.1.18.10)$$

$$= O(\log n) \quad (1.1.18.11)$$

This logarithmic performance bound relies entirely on executing the  $O(1)$  constant-time midpoint calculation, restricting its use strictly to contiguous memory structures like arrays, rendering it incompatible with linked lists.

1.1.19

$$A = I_n + xx^\top \quad (1.1.19.1)$$

Given the condition  $x^\top x = a$ , we can determine the eigenvalues by evaluating the standard eigenvalue equation  $Av = \lambda v$  across the two geometric subspaces defined by  $x$ .

For an eigenvector parallel to  $x$  (let  $v = x$ ):

$$Ax = (I_n + xx^\top)x \quad (1.1.19.2)$$

$$= x + x(x^\top x) \quad (1.1.19.3)$$

$$= x + x(a) \quad (1.1.19.4)$$

$$= (1 + a)x \quad (1.1.19.5)$$

yielding a single eigenvalue of  $\lambda_1 = 1 + a$ .

For any eigenvector  $v$  orthogonal to  $x$  (meaning  $x^\top v = 0$ ):

$$Av = (I_n + xx^\top)v \quad (1.1.19.6)$$

$$= v + x(x^\top v) \quad (1.1.19.7)$$

$$= v + x(0) \quad (1.1.19.8)$$

$$= 1v \quad (1.1.19.9)$$

yielding an eigenvalue of  $\lambda_2 = 1$  with multiplicity  $n - 1$ .

To derive the corresponding eigenvalues for the inverse matrix  $A^{-1}$ , we use the defining property of eigenvalues  $Av = \lambda v$  and left-multiply both sides by  $A^{-1}$ :

$$A^{-1}(Av) = A^{-1}(\lambda v) \quad (1.1.19.10)$$

$$(A^{-1}A)v = \lambda(A^{-1}v) \quad (1.1.19.11)$$

$$I_n v = \lambda A^{-1}v \quad (1.1.19.12)$$

$$v = \lambda A^{-1}v \quad (1.1.19.13)$$

$$A^{-1}v = \frac{1}{\lambda}v \quad (1.1.19.14)$$

This proves that any eigenvector  $v$  of  $A$  is also an eigenvector of  $A^{-1}$ , and its corresponding eigenvalue is exactly the reciprocal  $\frac{1}{\lambda}$ . Substituting our previously derived values, the eigenvalues of  $A^{-1}$  are  $\frac{1}{1+a}$  and 1.

### 1.1.20 Insertion Sort Inversions

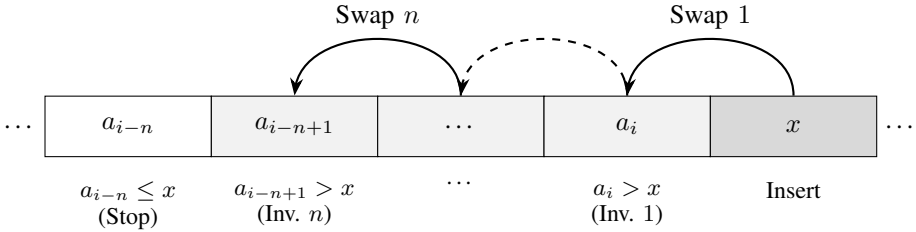


Fig. 1.1.20.1: Pictorial representation of Insertion Sort executing exactly  $n$  swaps for  $n$  inversions

$$\text{Swaps} = \text{Number of Inversions} \quad (1.1.20.1)$$

**Solution:** Insertion sort executes exactly one swap per inverted pair. Therefore, for the array to achieve a completely sorted state using exactly  $n$  swaps, the newly inserted element  $x$  must be strictly smaller than exactly  $n$  preceding elements.

### 1.1.21 Bayes' Theorem:

$$Pr(E_2|W) = \frac{Pr(E_2) Pr(W|E_2)}{\sum_i Pr(E_i) Pr(W|E_i)} \quad (1.1.21.1)$$

### 1.1.22 Asymptotic Radical Limits

$$\lim_{t \rightarrow +\infty} \left( \sqrt{t^2 + at} - t \right) = \frac{a}{2} \quad (1.1.22.1)$$

**Solution:** Multiplying by the conjugate:

$$\lim_{t \rightarrow +\infty} \frac{(t^2 + at) - t^2}{\sqrt{t^2 + at} + t} = \lim_{t \rightarrow +\infty} \frac{a}{\sqrt{1 + \frac{a}{t}} + 1} = \frac{a}{2} \quad (1.1.22.2)$$

### 1.1.23 Least Squares Regression Through Origin

$$w^* = \frac{\sum x_i y_i}{\sum x_i^2} \quad (1.1.23.1)$$

**Solution:** For a linear model constrained to pass strictly through the origin, the predicted value is  $\hat{y}_i = wx_i$ . The objective function  $S(w)$  represents the Sum of Squared Residuals (SSR), which measures the total squared penalty between the actual observed values  $y_i$  and the model's predictions  $\hat{y}_i$ :

$$S(w) = \sum (y_i - \hat{y}_i)^2 \quad (1.1.23.2)$$

$$= \sum (y_i - wx_i)^2 \quad (1.1.23.3)$$

To determine the optimal parameter  $w^*$  that minimizes this cumulative error, evaluate the derivative of  $S(w)$  with respect to  $w$  and equate it to zero to solve for the global minimum symbolically:

$$\frac{dS}{dw} = -2 \sum x_i (y_i - wx_i) = 0 \quad (1.1.23.4)$$

$$\sum x_i y_i - w \sum x_i^2 = 0 \quad (1.1.23.5)$$

$$w^* = \frac{\sum x_i y_i}{\sum x_i^2} \quad (1.1.23.6)$$

### 1.1.24 Bayes Optimal Error Rate

$$Pr(\text{error} | \mathbf{x}) = \frac{\min_k Pr(\mathbf{x} | y_k) Pr(y_k)}{Pr(\mathbf{x})} \quad (1.1.24.1)$$

### 1.1.25 Moments of Standard Normal

$$Z \sim \mathcal{N}(0, 1) \implies E[Z^2] = 1, E[Z^4] = 3 \quad (1.1.25.1)$$

$$var(Y) = E[Y^2] - (E[Y])^2 \quad (1.1.25.2)$$

### 1.1.26 Rank of Involutory Matrices

$$\mathbf{A}^3 = \mathbf{A} \implies \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^2) \quad (1.1.26.1)$$

**Solution:**

Let  $\mathbf{z} \in \text{Col}(\mathbf{A}^2)$ . By definition, there exists a vector  $\mathbf{v}$  such that:

$$\mathbf{z} = \mathbf{A}^2 \mathbf{v} \quad (1.1.26.2)$$

$$= \mathbf{A}(\mathbf{A} \mathbf{v}) \quad (1.1.26.3)$$

Letting  $\mathbf{u} = \mathbf{A} \mathbf{v}$ , we have  $\mathbf{z} = \mathbf{A} \mathbf{u}$ , which dictates  $\mathbf{z} \in \text{Col}(\mathbf{A})$ . Thus:

$$\text{Col}(\mathbf{A}^2) \subseteq \text{Col}(\mathbf{A}) \implies \text{rank}(\mathbf{A}^2) \leq \text{rank}(\mathbf{A}) \quad (1.1.26.4)$$

Conversely, let  $\mathbf{y} \in \text{Col}(\mathbf{A})$ . Then there exists a vector  $\mathbf{x}$  such that:

$$\mathbf{y} = \mathbf{A}\mathbf{x} \quad (1.1.26.5)$$

$$= \mathbf{A}^3\mathbf{x} \quad (\text{given } \mathbf{A} = \mathbf{A}^3) \quad (1.1.26.6)$$

$$= \mathbf{A}^2(\mathbf{A}\mathbf{x}) \quad (1.1.26.7)$$

This demonstrates  $\mathbf{y} \in \text{Col}(\mathbf{A}^2)$ , establishing the reverse inclusion:

$$\text{Col}(\mathbf{A}) \subseteq \text{Col}(\mathbf{A}^2) \implies \text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{A}^2) \quad (1.1.26.8)$$

Consequently, the ranks must be equal:

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^2) \quad (1.1.26.9)$$

1.1.27

$$\mathbf{A} = \mathbf{X}^\top \mathbf{X} \implies \mathbf{z}^\top \mathbf{A} \mathbf{z} = \|\mathbf{X}\mathbf{z}\|^2 \quad (1.1.27.1)$$

If the columns of  $\mathbf{X}$  are linearly independent,  $\mathbf{X}\mathbf{z} = \mathbf{0} \iff \mathbf{z} = \mathbf{0}$ . Thus,  $\|\mathbf{X}\mathbf{z}\|^2 > 0$  for all non-zero  $\mathbf{z}$ , rendering  $\mathbf{A}$  positive definite and invertible.

1.1.28 Interval Probabilities from CDF

$$\Pr(X^2 \leq c^2) = F_X(c) - F_X(-c) \quad (1.1.28.1)$$

**Solution:** For any constant  $c > 0$ , the quadratic inequality  $X^2 \leq c^2$  algebraically resolves to a bounded symmetric interval. The probability of this event is evaluated by separating the interval bounds:

$$\Pr(X^2 \leq c^2) = \Pr(-c \leq X \leq c) \quad (1.1.28.2)$$

$$= \Pr(X > -c \text{ and } X \leq c) \quad (1.1.28.3)$$

$$= \Pr(X \leq c) - \Pr(X \leq -c) \quad (1.1.28.4)$$

Substituting the fundamental definition of the Cumulative Distribution Function,  $F_X(x) = \Pr(X \leq x)$  (assuming a continuous distribution where  $\Pr(X = -c) = 0$ ), yields the final expression:

$$\Pr(X^2 \leq c^2) = F_X(c) - F_X(-c) \quad (1.1.28.5)$$

1.1.29 Central Limit Theorem Approximation for Binomial Sums

$$Z = \frac{Y - n\theta}{\sqrt{n\theta(1-\theta)}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (1.1.29.1)$$

For independent variables  $X_i \sim \text{Bernoulli}(\theta)$ , the sum  $Y$ , its mean  $\mu$ , and standard deviation  $\sigma$  are defined as:

$$Y = \sum_{i=1}^n X_i \quad (1.1.29.2)$$

$$\mu = n\theta \quad (1.1.29.3)$$

$$\sigma = \sqrt{n\theta(1-\theta)} \quad (1.1.29.4)$$

Standardizing the interval  $[a, b]$  maps the boundaries to the standard normal distribution  $Z$ . The probability evaluates to the difference of the standard normal Cumulative Distribution Function (CDF), denoted by  $\phi(x)$ , at these bounds:

$$P(a \leq Y \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) \quad (1.1.29.5)$$

$$= \phi\left(\frac{b - n\theta}{\sqrt{n\theta(1-\theta)}}\right) - \phi\left(\frac{a - n\theta}{\sqrt{n\theta(1-\theta)}}\right) \quad (1.1.29.6)$$

### 1.1.30

$$\Pr(\lfloor X \rfloor = \ell) = e^{-\lambda\ell} (1 - e^{-\lambda}) \quad (1.1.30.1)$$

**Solution:** Let  $X$  be an exponentially distributed random variable with rate parameter  $\lambda = \frac{1}{\mu}$ . Its probability density function is  $f_X(x) = \lambda e^{-\lambda x}$ .

The event  $Y = \lfloor X \rfloor = \ell$  strictly occurs when  $X$  falls within the bounded interval  $[\ell, \ell + 1)$ . The probability is derived by integrating the PDF over these boundaries:

$$P(Y = \ell) = P(\ell \leq X < \ell + 1) \quad (1.1.30.2)$$

$$= \int_{\ell}^{\ell+1} \lambda e^{-\lambda x} dx \quad (1.1.30.3)$$

$$= [-e^{-\lambda x}]_{\ell}^{\ell+1} \quad (1.1.30.4)$$

$$= e^{-\lambda\ell} - e^{-\lambda(\ell+1)} \quad (1.1.30.5)$$

$$= e^{-\lambda\ell} (1 - e^{-\lambda}) \quad (1.1.30.6)$$

By substituting  $q = e^{-\lambda}$ , the exact geometric probability mass function emerges:

$$P(Y = \ell) = q^{\ell} (1 - q) \quad (1.1.30.7)$$

### 1.1.31 Probability of At Least One Success via Z-Transform

$$M_X(z) = (q + pz^{-1})^n \quad (1.1.31.1)$$

Let  $X = \sum_{i=1}^n X_i$  represent the total sum of  $n$  independent trials, where each  $X_i \sim \text{Bernoulli}(p)$ . The Z-transform for a single trial is  $M_{X_i}(z) = q + pz^{-1}$ , and the aggregate transform is:

$$M_X(z) = (q + pz^{-1})^n \quad (1.1.31.2)$$

Expanding this polynomial yields the probability mass function  $p_X(k)$  as the coefficient of the  $z^{-k}$  term. The probability of zero successes ( $k = 0$ ) corresponds to the constant term  $q^n$ . Applying the complement rule to find the probability of at least one success:

$$P(X \geq 1) = 1 - p_X(0) \quad (1.1.31.3)$$

$$= 1 - q^n \quad (1.1.31.4)$$

## 1.1.32

$$f'(x) = 0 \text{ at local extrema} \quad (1.1.32.1)$$

## 1.1.33 Properties of Orthogonal Projection Matrices

$$\mathbf{A} = \sum_{i=1}^k \mathbf{x}_i \mathbf{x}_i^\top \implies \mathbf{A}^2 = \mathbf{A} \quad (1.1.33.1)$$

**Solution:** For an orthonormal set of vectors where the inner product  $\mathbf{x}_i^\top \mathbf{x}_j = \delta_{ij}$ :

$$\mathbf{A}^2 = \left( \sum_{i=1}^k \mathbf{x}_i \mathbf{x}_i^\top \right) \left( \sum_{j=1}^k \mathbf{x}_j \mathbf{x}_j^\top \right) \quad (1.1.33.2)$$

$$= \sum_{i=1}^k \sum_{j=1}^k \mathbf{x}_i (\mathbf{x}_i^\top \mathbf{x}_j) \mathbf{x}_j^\top \quad (1.1.33.3)$$

$$= \sum_{i=1}^k \mathbf{x}_i (1) \mathbf{x}_i^\top \quad (1.1.33.4)$$

$$= \mathbf{A} \quad (1.1.33.5)$$

## 1.1.34 Isometric Norm Preservation

$$\|\mathbf{Ax}\|^2 = \|\mathbf{x}\|^2 \iff \mathbf{A}^\top \mathbf{A} = \mathbf{I} \quad (1.1.34.1)$$

**Solution:** By the definition of the Euclidean norm in terms of the standard inner product, evaluate both sides for an arbitrary vector  $\mathbf{x} \in \mathbb{R}^n$ :

$$\|\mathbf{Ax}\|^2 = (\mathbf{Ax})^\top (\mathbf{Ax}) = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} \quad (1.1.34.2)$$

$$\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{x} = \mathbf{x}^\top \mathbf{I} \mathbf{x} \quad (1.1.34.3)$$

Equating the two expanded forms for all  $\mathbf{x}$ :

$$\mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{I} \mathbf{x} \quad (1.1.34.4)$$

$$\mathbf{x}^\top (\mathbf{A}^\top \mathbf{A} - \mathbf{I}) \mathbf{x} = 0 \quad (1.1.34.5)$$

The matrix  $(\mathbf{A}^\top \mathbf{A} - \mathbf{I})$  is inherently symmetric. If a quadratic form of a symmetric matrix evaluates to zero for all possible vectors  $\mathbf{x}$ , the matrix must be the zero matrix. Therefore:

$$\mathbf{A}^\top \mathbf{A} - \mathbf{I} = \mathbf{0} \implies \mathbf{A}^\top \mathbf{A} = \mathbf{I} \quad (1.1.34.6)$$

## 1.1.35 Properties of Sample Proportion

$$E[\hat{p}] = p \quad (1.1.35.1)$$

$$\text{var}(\hat{p}) = \frac{p(1-p)}{n} \quad (1.1.35.2)$$

**Solution:** Let  $X = \sum_{i=1}^n X_i$  represent the total number of heads. Because  $X_i$  are

independent Bernoulli random variables, their sum follows a Binomial distribution  $X \sim B(n, p)$ .

From the defined moments of a Binomial distribution:

$$E(X) = np \quad (1.1.35.3)$$

$$\text{var}(X) = npq = np(1 - p) \quad (1.1.35.4)$$

The sample proportion is given as  $\hat{p} = \frac{X}{n}$ . Evaluating the expectation yields:

$$E[\hat{p}] = E\left[\frac{X}{n}\right] \quad (1.1.35.5)$$

$$= \frac{1}{n} E(X) \quad (1.1.35.6)$$

$$= \frac{1}{n} (np) = p \quad (1.1.35.7)$$

To evaluate the variance, apply the fundamental variance scaling property  $\text{var}(aX) = a^2 \text{var}(X)$  established in the text:

$$\text{var}(\hat{p}) = \text{var}\left(\frac{X}{n}\right) \quad (1.1.35.8)$$

$$= \frac{1}{n^2} \text{var}(X) \quad (1.1.35.9)$$

$$= \frac{1}{n^2} [np(1 - p)] \quad (1.1.35.10)$$

$$= \frac{p(1 - p)}{n} \quad (1.1.35.11)$$

Because  $n$  is in the denominator of the resulting expression, the variance of  $\hat{p}$  strictly decreases as the total number of independent coin tosses  $n$  increases. Therefore, statements (A) and (C) are correct.

### 1.1.36 Shortest Path Metric

$$d(u, v) \leq d(u, w) + d(w, v) \quad (1.1.36.1)$$

Fake edges violating the triangle inequality are identifiable by calculating the minimum spanning distances directly.

### 1.1.37 Lipschitz Continuity with Exponent $> 1$

$$|f(x) - f(y)| \leq K |x - y|^\alpha \implies f'(x) = 0 \quad (1.1.37.1)$$

**Solution:** Dividing by  $|x - y|$  and taking the limit  $y \rightarrow x$  for  $\alpha > 1$  yields a derivative of zero, enforcing a constant function.

### 1.1.38 Projected Variance and Principal Components

$$\mathbf{C} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}^{(i)} \mathbf{x}^{(i)\top} \quad (\text{Covariance Matrix for centered data}) \quad (1.1.38.1)$$

$$\sigma_{\mathbf{u}}^2 = \frac{1}{n} \sum_{i=1}^n \left( \mathbf{u}^\top \mathbf{x}^{(i)} \right)^2 = \mathbf{u}^\top \mathbf{C} \mathbf{u} \quad (\text{Variance along direction } \mathbf{u}) \quad (1.1.38.2)$$

$$\max_{\mathbf{u}^\top \mathbf{u}=1} \mathbf{u}^\top \mathbf{C} \mathbf{u} = \lambda_{\max} \quad (\text{Rayleigh Quotient / Maximum Variance}) \quad (1.1.38.3)$$

### 1.1.39 Expectation of Indicator Sums

$$E \left[ \sum_{i=1}^n I_i \right] = \sum_{i=1}^n E[I_i] = np \quad (1.1.39.1)$$

**Solution:** Linearity of expectation holds uniformly irrespective of variable dependence or independence.

### 1.1.40 Relational Division

$R$  (**Student**, **Skill**)

| A       | B      |
|---------|--------|
| Alice   | SQL    |
| Alice   | Java   |
| Alice   | Python |
| Bob     | SQL    |
| Bob     | Python |
| Charlie | SQL    |
| Charlie | Java   |
| Charlie | Python |

$S$  (**Required**)

| B      |
|--------|
| SQL    |
| Java   |
| Python |

$\div$

$=$

$R \div S$

| A       |
|---------|
| Alice   |
| Charlie |

$$R \div S = \pi_A(R) - \pi_A((\pi_A(R) \times S) - R) \quad (1.1.40.1)$$

### 1.1.41 Depth-First Search (DFS) Traversal Sequence

$$V = \{0, 1, \dots, n\}, \quad E = \{(i, j) : 0 < j - i \leq 2\} \quad (1.1.41.1)$$

**Solution:** Decreasing adjacency order restricts the neighbor queue to  $Adj(i) = [i + 2, i + 1]$ . A DFS from 0 greedily traverses even vertices forward via  $i + 2$  edges, then odd vertices during backtracking via  $i + 1$  edges. For an even boundary  $n$ , the

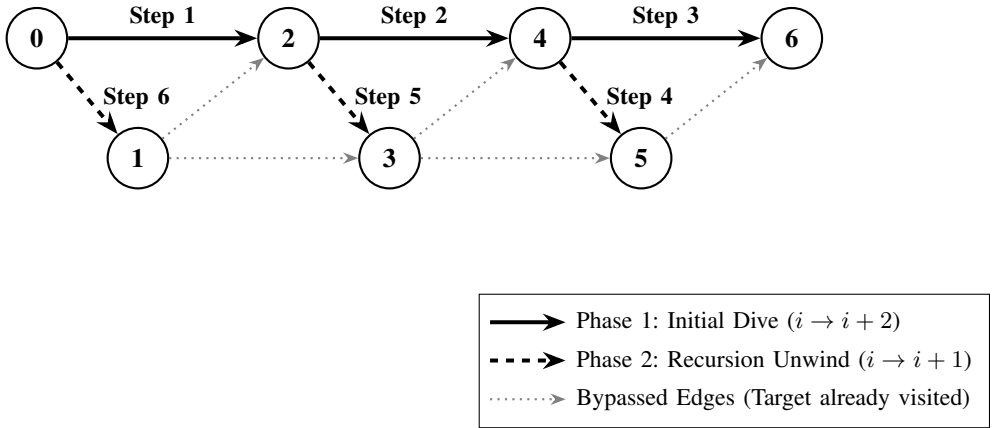


Fig. 1.1.41.1: DFS Traversal for  $n = 6$ . Decreasing adjacency priority forces the DFS to exhaust all even vertices forward via  $i + 2$  edges. It then unwinds sequentially, mapping the bypassed odd vertices via the  $i + 1$  edges in reverse chronological order.

vertex  $D(k)$  discovered at step  $k \in \{0, \dots, n\}$  is:

$$D(k) = \begin{cases} 2k & 0 \leq k \leq n/2 \quad (\text{Forward}) \\ 2n - 2k + 1 & n/2 < k \leq n \quad (\text{Backtrack}) \end{cases} \quad (1.1.41.2)$$

To find the number of vertices discovered chronologically after a target  $v$ , solve  $D(k_v) = v$  for its discovery index  $k_v$ . Out of  $n + 1$  total vertices, the remaining unvisited count evaluates to:

$$\text{Remaining Discoveries} = n - k_v \quad (1.1.41.3)$$

### B. Problems

- 1.2.1 A  $4 \times 4$  digital image has pixel intensities ( $U$ ) as shown in the figure. The number of pixels with  $U \leq 4$  is: (DA 2025)

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 0 | 2 |
| 4 | 7 | 3 | 3 |
| 5 | 5 | 4 | 4 |
| 6 | 7 | 3 | 2 |

Fig. 1.2.1.1: Pixel Grid

a) 3

b) 8

c) 11

d) 9

**Solution:** From (1.1.1.1) and (1.1.1.2), for threshold  $c = 4$ :

$$N = \sum_{i=1}^4 \sum_{j=1}^4 \mathbb{I}(U_{ij}) \quad (1.2.1.1)$$

$$= 11 \quad (1.2.1.2)$$

1.2.2 A rectangle has a length  $L$  and a width  $W$ , where  $L > W$ . If the width,  $W$ , is increased by 10%, which one of the following statements is correct for all values of  $L$  and  $W$ ? (DA 2025)

- a) Perimeter increases by 10%.
- b) Length of the diagonals increases by 10%.
- c) Area increases by 10%.
- d) The rectangle becomes a square.

**Solution:** From (1.1.2.1), with width scaled by  $\alpha = 1.10$ :

$$A' = (1 + 0.10) A \quad (1.2.2.1)$$

$$= 1.10A \quad (1.2.2.2)$$

1.2.3 Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let  $a_1$  and  $a_2$  (in years) denote the ages of the brothers and  $a_1 < a_2$ . Which one of the following statements is correct about their age on the day when they attain the same weight? (DA 2025)

- a)  $a_1 < a_2 < 50$
- b)  $a_1 < 50 < a_2$
- c)  $50 < a_1 < a_2$
- d) Either  $a_1 = 50$  or  $a_2 = 50$

**Solution:** From (1.1.3.1), for peak  $c = 50$ :

$$f(a_1) = f(a_2) \quad (1.2.3.1)$$

$$\implies a_1 < 50 < a_2 \quad (1.2.3.2)$$

1.2.4 If a real variable  $x$  satisfies  $3^{x^2} = 27 \times 9^x$ , then the value of  $\frac{2^{x^2}}{(2^x)^2}$  is: (DA 2025)

a)  $2^{-1}$ b)  $2^0$ c)  $2^3$ d)  $2^{15}$ 

**Solution:** From (1.1.4.1):

$$3^{x^2} = 3^3 \cdot (3^2)^x \quad (1.2.4.1)$$

$$= 3^{2x+3} \quad (1.2.4.2)$$

$$\implies x^2 - 2x = 3 \quad (1.2.4.3)$$

Applying (1.1.4.1):

$$\frac{2^{x^2}}{(2^x)^2} = \frac{2^{x^2}}{2^{2x}} \quad (1.2.4.4)$$

$$= 2^{x^2-2x} \quad (1.2.4.5)$$

$$= 2^3 \quad (1.2.4.6)$$

1.2.5 Suppose  $X$  and  $Y$  are random variables. The conditional expectation of  $X$  given  $Y$  is denoted by  $E[X | Y]$ . Then  $E[E[X | Y]]$  equals (DA 2025)

- a)  $E[X | Y]$       b)  $\frac{E[X]}{E[Y]}$       c)  $E[X]$       d)  $E[Y]$

**Solution:** From (1.1.5.1) and (1.1.5.2):

$$E[E[X | Y]] = \sum_y E[X | Y = y] p_Y(y) \quad (1.2.5.1)$$

$$= \sum_y \sum_x x p_{X|Y}(x | y) p_Y(y) \quad (1.2.5.2)$$

$$= \sum_x x \sum_y p_{X,Y}(x, y) \quad (1.2.5.3)$$

$$= \sum_x x p_X(x) \quad (1.2.5.4)$$

$$= E[X] \quad (1.2.5.5)$$

1.2.6 The number of additions and multiplications involved in performing Gaussian elimination on any  $n \times n$  upper triangular matrix is of the order (DA 2025)

- a)  $O(n)$       b)  $O(n^2)$       c)  $O(n^3)$       d)  $O(n^4)$

**Solution:** From (1.1.6.1):

$$x_i = \frac{1}{U_{ii}} \left( b_i - \sum_{j=i+1}^n U_{ij} x_j \right) \quad (1.2.6.1)$$

For each row  $i$  ( $1 \leq i \leq n$ ), the back-substitution step requires:

- Multiplications:  $n - i$
- Additions/Subtractions:  $n - i$
- Divisions: 1

Summing these operations across all  $n$  rows:

$$\text{Multiplications} = \sum_{i=1}^n (n-i) = \frac{n(n-1)}{2} \quad (1.2.6.2)$$

$$\text{Additions} = \sum_{i=1}^n (n-i) = \frac{n(n-1)}{2} \quad (1.2.6.3)$$

$$\text{Divisions} = \sum_{i=1}^n 1 = n \quad (1.2.6.4)$$

The total sum of all arithmetic operations evaluates to:

$$\text{Total Operations} = \frac{n(n-1)}{2} + \frac{n(n-1)}{2} + n = n^2 \quad (1.2.6.5)$$

$$= O(n^2) \quad (1.2.6.6)$$

1.2.7 The sum of the elements in each row of  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is 1. If  $\mathbf{B} = \mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}$ , which one of the following statements is correct (for  $\mathbf{x} \in \mathbb{R}^n$ )? (DA 2025)

- a) The equation  $\mathbf{B}\mathbf{x} = \mathbf{0}$  has no solution
- b) The equation  $\mathbf{B}\mathbf{x} = \mathbf{0}$  has exactly two solutions
- c) The equation  $\mathbf{B}\mathbf{x} = \mathbf{0}$  has infinitely many solutions
- d) The equation  $\mathbf{B}\mathbf{x} = \mathbf{0}$  has a unique solution

**Solution:** From (1.1.7.1),  $\mathbf{A}\mathbf{1} = \mathbf{1}$ . For  $\mathbf{B} = \mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}$ :

$$\mathbf{B}\mathbf{1} = (\mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A})\mathbf{1} \quad (1.2.7.1)$$

$$= \mathbf{A}^3\mathbf{1} - 2\mathbf{A}^2\mathbf{1} + \mathbf{A}\mathbf{1} \quad (1.2.7.2)$$

$$= \mathbf{1} - 2\mathbf{1} + \mathbf{1} \quad (1.2.7.3)$$

$$= \mathbf{0} \quad (1.2.7.4)$$

Thus,  $\lambda = 0$  is an eigenvalue of  $\mathbf{B}$ , implying  $|\mathbf{B}| = 0$ , which yields infinitely many solutions.

1.2.8 Let  $f(x) = \frac{e^x - e^{-x}}{2}$ ,  $x \in \mathbb{R}$ . Let  $f^{(k)}(a)$  denote the  $k^{\text{th}}$  derivative of  $f$  evaluated at  $a$ . What is the value of  $f^{(10)}(0)$ ? (Note: ! denotes factorial) (DA 2025)

- a) 0
- b) 1
- c)  $\frac{1}{10!}$
- d)  $\frac{2}{10!}$

**Solution:** From (1.1.8.1):

$$f(x) = \sinh(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} \quad (1.2.8.1)$$

By definition, the general Maclaurin series coefficient for  $x^n$  is given by  $\frac{f^{(n)}(0)}{n!}$ .

Since the series contains only odd powers of  $x$ , the coefficient of  $x^{10}$  is zero:

$$\frac{f^{(10)}(0)}{10!} = 0 \quad (1.2.8.2)$$

$$\implies f^{(10)}(0) = 0 \quad (1.2.8.3)$$

1.2.9 Consider the following three relations:

Car (model, year, serial, color)

Make (maker, model)

Own (owner, serial)

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; (owner, serial) together form key for Own. ( $\bowtie$  denotes natural join)

$$\pi_{\text{owner}} (\text{Own} \bowtie (\sigma_{\text{color}='red'} (\text{Car} \bowtie (\sigma_{\text{maker}='ABC'} \text{Make})))) \quad (1.2.9.1)$$

Which one of the following options describes what the above expression computes? (DA 2025)

- a) All owners of a red car, a car made by ABC, or a red car made by ABC
- b) All owners of more than one car, where at least one car is red and made by ABC
- c) All owners of a red car made by ABC
- d) All red cars made by ABC

**Solution:** From (1.1.9.1),  $\sigma_{\text{color}='red'} (\text{Car} \bowtie (\sigma_{\text{maker}='ABC'} \text{Make}))$  filters red cars made by ABC. Joining with Own and projecting via  $\pi_{\text{owner}}$  selects:

$$\pi_{\text{owner}} (\text{Own} \bowtie (\dots)) \quad (1.2.9.2)$$

which corresponds to all owners of a red car made by ABC.

1.2.10 Consider a hash table of size 10 with indices  $\{0, 1, \dots, 9\}$ , with the hash function

$$h(x) = 3x \pmod{10} \quad (1.2.10.1)$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: 1, 4, 5, 6, 14, 15. The indices where the keys 14 and 15 are stored are, respectively (DA 2025)

- a) 2 and 5
- b) 2 and 6
- c) 4 and 5
- d) 4 and 6

**Solution:** From (1.1.10.1),  $h(k, i) = (3k + i) \pmod{10}$ :

$$k = 1 \implies h(1, 0) = 3 \quad (1.2.10.2)$$

$$k = 4 \implies h(4, 0) = 2 \quad (1.2.10.3)$$

$$k = 5 \implies h(5, 0) = 5 \quad (1.2.10.4)$$

$$k = 6 \implies h(6, 0) = 8 \quad (1.2.10.5)$$

$$k = 14 \implies h(14, 0) = 2 \quad (1.2.10.6)$$

$$\implies h(14, 1) = 3 \quad (1.2.10.7)$$

$$\implies h(14, 2) = 4 \quad (1.2.10.8)$$

$$k = 15 \implies h(15, 0) = 5 \quad (1.2.10.9)$$

$$\implies h(15, 1) = 6 \quad (1.2.10.10)$$

1.2.11 Let  $X$  be a continuous random variable whose cumulative distribution function (CDF)  $F_X(x)$ , for some  $t$ , is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \frac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases} \quad (1.2.11.1)$$

If the median of  $X$  is 3, then what is the value of  $t$ ? (DA 2025)

- a) 2                      b) 1                      c) -1                      d) 0

**Solution:** From (1.1.11.3), the median  $m = 3$  satisfies:

$$F_X(3) = 0.5 \quad (1.2.11.2)$$

$$\frac{3-t}{4-t} = 0.5 \quad (1.2.11.3)$$

$$3-t = 2 - 0.5t \quad (1.2.11.4)$$

$$\implies t = 2 \quad (1.2.11.5)$$

1.2.12 Let  $X = aZ + b$ , where  $Z$  is a standard normal random variable, and  $a, b$  are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4, \quad (1.2.12.1)$$

where  $E[X]$  denotes the expectation of random variable  $X$ . The values of  $a, b$  are: (DA 2025)

- a)  $a = -2, b = 1$                       c)  $a = -2, b = -1$   
b)  $a = 2, b = -1$                       d)  $a = 1, b = 1$

**Solution:** From (1.1.13.1) and (1.1.13.5), for  $X = aZ + b$ :

$$E[X] = aE[Z] + b \quad (1.2.12.2)$$

$$\implies b = 1 \quad (1.2.12.3)$$

From (1.1.13.2):

$$E[(X - E[X])Z] = E[aZ^2] \quad (1.2.12.4)$$

$$= a(1) \quad (1.2.12.5)$$

$$\implies a = -2 \quad (1.2.12.6)$$

1.2.13 It is given that  $P(X \geq 2) = 0.25$  for an exponentially distributed random variable  $X$  with  $E[X] = \frac{1}{\lambda}$ , where  $E[X]$  denotes the expectation of  $X$ . What is the value of  $\lambda$ ? (ln denotes natural logarithm) (DA 2025)

- a)  $\ln 2$                       b)  $\ln 4$                       c)  $\ln 3$                       d)  $\ln 0.25$

**Solution:** From (1.1.14.1):

$$\Pr(X \geq 2) = e^{-2\lambda} = 0.25 = \frac{1}{4} \quad (1.2.13.1)$$

$$\implies -2\lambda = \ln\left(\frac{1}{4}\right) = -\ln 4 \quad (1.2.13.2)$$

$$\implies 2\lambda = 2\ln 2 \quad (1.2.13.3)$$

$$\implies \lambda = \ln 2 \quad (1.2.13.4)$$

1.2.14 Consider the following Python declarations of two lists.

A = [1, 2, 3]  
B = [4, 5, 6]

Which one of the following statements results in  $A = [1, 2, 3, 4, 5, 6]$ ? (DA 2025)

- a) A.extend(B)              b) A.append(B)              c) A.update(B)              d) A.insert(B)

**Solution:** From (1.1.15.1):

$$A.\text{extend}(B) \implies A \leftarrow [1, 2, 3, 4, 5, 6] \quad (1.2.14.1)$$

1.2.15 Consider two functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow (1, \infty)$ . Both functions are differentiable at a point  $c$ . Which of the following functions is/are ALWAYS differentiable at  $c$ ? The symbol  $\cdot$  denotes product and the symbol  $\circ$  denotes composition of functions. (DA 2025)

- a)  $f \pm g$                       b)  $f \cdot g$                       c)  $\frac{f}{g}$                       d)  $f \circ g + g \circ f$

**Solution:** From (1.1.16.1), the product  $f \cdot g$ , sum  $f \pm g$ , and quotient  $f/g$  (since  $g(c) > 1 \neq 0$ ) are strictly differentiable at  $c$ .  $f \circ g$  requires differentiability of  $f$  at  $g(c)$ , which is not guaranteed.

1.2.16 Which of the following statements is/are correct? (DA 2025)

- a)  $\mathbb{R}^n$  has a unique set of orthonormal basis vectors  
b)  $\mathbb{R}^n$  does not have a unique set of orthonormal basis vectors

c) Linearly independent vectors in  $\mathbb{R}^n$  are orthonormal

d) Orthonormal vectors in  $\mathbb{R}^n$  are linearly independent

**Solution:** From (1.1.17.1),  $\mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij}$  guarantees linear independence. Orthonormal bases in  $\mathbb{R}^n$  are non-unique under spatial rotation.

1.2.17 For which of the following inputs does binary search take time  $O(\log n)$  in the worst case? (DA 2025)

a) An array of  $n$  integers in any order

b) A linked list of  $n$  integers in any order

c) An array of  $n$  integers in increasing order

d) A linked list of  $n$  integers in increasing order

**Solution:** From (1.1.18.1) and (1.1.18.2):

$$T(n) = T(n/2) + O(1) \quad (1.2.17.1)$$

$$= O(\log n) \quad (1.2.17.2)$$

which requires  $O(1)$  random access to the median element, satisfied exclusively by an array of  $n$  integers in increasing order.

1.2.18 Let  $\mathbf{A} = \mathbf{I}_n + \mathbf{x}\mathbf{x}^\top$  where  $\mathbf{I}_n$  is the  $n \times n$  identity matrix and  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{x}^\top \mathbf{x} = 1$ . Which of the following options is/are correct? (DA 2025)

a) Rank of  $\mathbf{A}$  is  $n$

b)  $\mathbf{A}$  is invertible

c) 0 is an eigenvalue of  $\mathbf{A}$

d)  $\mathbf{A}^{-1}$  has a negative eigenvalue

**Solution:** From (1.1.19.1), evaluating the eigenvalues for  $\mathbf{x}^\top \mathbf{x} = 1$ :

$$\mathbf{A}\mathbf{x} = \mathbf{x} + \mathbf{x}(\mathbf{x}^\top \mathbf{x}) \quad (1.2.18.1)$$

$$= 2\mathbf{x} \quad (1.2.18.2)$$

$$\implies \lambda_1 = 2 \quad (1.2.18.3)$$

$$\mathbf{A}\mathbf{v} = \mathbf{v} + \mathbf{x}(\mathbf{x}^\top \mathbf{v}) \quad (1.2.18.4)$$

$$= 1\mathbf{v} \quad (1.2.18.5)$$

$$\implies \lambda_{2,\dots,n} = 1 \quad \text{for } \mathbf{x}^\top \mathbf{v} = 0 \quad (1.2.18.6)$$

Since all  $\lambda_i > 0$ , rank( $\mathbf{A}$ ) =  $n$  and  $\mathbf{A}$  is invertible.

1.2.19 Suppose that insertion sort is applied to the array  $[1, 3, 5, 7, 9, 11, x, 15, 13]$  and it takes exactly two swaps to sort the array. Select all possible values of  $x$ . (DA 2025)

a) 10

b) 12

c) 14

d) 16

**Solution:** From (1.1.20.1):

$$\text{Swaps} = \text{Inversions} = 2 \quad (1.2.19.1)$$

The pair (15, 13) accounts for 1 inversion. Therefore,  $x$  contributes exactly 1 inversion

with  $[1, 3, 5, 7, 9, 11]$ :

$$9 \leq x < 11 \quad (1.2.19.2)$$

$$\implies x = 10 \quad (1.2.19.3)$$

1.2.20 There are three boxes containing white balls and black balls.

- Box-1 contains 2 black and 1 white balls.
- Box-2 contains 1 black and 2 white balls.
- Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is  $\frac{1}{2}$ , Box-2 is  $\frac{1}{6}$ , and Box-3 is  $\frac{1}{3}$ . A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is \_\_\_\_\_.

(DA 2025)

**Solution:** From (1.1.21.1):

$$Pr(E_2 | W) = \frac{Pr(E_2) Pr(W | E_2)}{\sum_{i=1}^3 Pr(E_i) Pr(W | E_i)} \quad (1.2.20.1)$$

$$= \frac{\frac{1}{6} \cdot \frac{2}{3}}{\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{3}{6}} \quad (1.2.20.2)$$

$$= \frac{\frac{1}{9}}{\frac{1}{6} + \frac{1}{9} + \frac{1}{6}} \quad (1.2.20.3)$$

$$= \frac{\frac{1}{9}}{\frac{4}{9}} \quad (1.2.20.4)$$

$$= \frac{1}{4} = 0.25 \quad (1.2.20.5)$$

1.2.21  $\lim_{t \rightarrow +\infty} \sqrt{t^2 + t} - t = \underline{\hspace{2cm}}.$  (DA 2025)

**Solution:** From (1.1.22.1), with  $a = 1$ :

$$\lim_{t \rightarrow +\infty} \left( \sqrt{t^2 + t} - t \right) = \frac{1}{2} = 0.5 \quad (1.2.21.1)$$

1.2.22 On a relation named Loan of a bank:

| loan_number | branch_name   | amount |
|-------------|---------------|--------|
| L11         | Banjara Hills | 90000  |
| L14         | Kondapur      | 50000  |
| L15         | SR Nagar      | 40000  |
| L22         | SR Nagar      | 25000  |
| L23         | Balanagar     | 80000  |
| L25         | Kondapur      | 70000  |
| L19         | SR Nagar      | 65000  |

Fig. 1.2.22.1

the following SQL query is executed.

```

SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
FROM Loan L2
WHERE L2.branch_name = 'SR Nagar');

```

The number of rows returned by the query is \_\_\_\_\_. (DA 2025)

- 1.2.23 Given data  $\{(-1, 1), (2, 5), (3, 5)\}$  of the form  $(x, y)$ , we fit a model  $y = wx$  using linear least-squares regression. The optimal value of  $w$  is \_\_\_\_\_. (DA 2025)

**Solution:** From (1.1.23.1):

$$w^* = \frac{\sum x_i y_i}{\sum x_i^2} \quad (1.2.23.1)$$

$$= \frac{(-1)(1) + (2)(5) + (3)(5)}{(-1)^2 + 2^2 + 3^2} \quad (1.2.23.2)$$

$$= \frac{24}{14} \quad (1.2.23.3)$$

$$= \frac{12}{7} \approx 1.714 \quad (1.2.23.4)$$

- 1.2.24 The naive Bayes classifier is used to solve a two-class classification problem with class-labels  $y_1, y_2$ . Suppose the prior probabilities are  $P(y_1) = \frac{1}{3}$  and  $P(y_2) = \frac{2}{3}$ . Assuming a discrete feature space with

$$P(\mathbf{x} | y_1) = \frac{3}{4} \quad \text{and} \quad P(\mathbf{x} | y_2) = \frac{1}{4} \quad (1.2.24.1)$$

for a specific feature vector  $\mathbf{x}$ . The probability of misclassifying  $\mathbf{x}$  is \_\_\_\_\_. (DA 2025)

**Solution:** From (1.1.24.1):

$$Pr(\text{error} | \mathbf{x}) = \frac{\min_k Pr(\mathbf{x} | y_k) Pr(y_k)}{Pr(\mathbf{x})} \quad (1.2.24.2)$$

$$= \frac{\min\left(\frac{3}{4} \cdot \frac{1}{3}, \frac{1}{4} \cdot \frac{2}{3}\right)}{\frac{3}{4} \cdot \frac{1}{3} + \frac{1}{4} \cdot \frac{2}{3}} \quad (1.2.24.3)$$

$$= \frac{\min\left(\frac{1}{4}, \frac{1}{6}\right)}{\frac{1}{4} + \frac{1}{6}} \quad (1.2.24.4)$$

$$= \frac{\frac{1}{6}}{\frac{5}{12}} \quad (1.2.24.5)$$

$$= \frac{2}{5} = 0.4 \quad (1.2.24.6)$$

- 1.2.25 Let  $Y = Z^2$ ,  $Z = \frac{X - \mu}{\sigma}$ , where  $X$  is a normal random variable with mean  $\mu$  and variance  $\sigma^2$ . The variance of  $Y$  is \_\_\_\_\_. (DA 2025)

a) 1

b) 2

c) 3

d) 4

**Solution:** From (1.1.25.2) and (1.1.25.1), with  $Z \sim \mathcal{N}(0, 1)$ :

$$\text{var}(Y) = E[Z^4] - (E[Z^2])^2 \quad (1.2.25.1)$$

$$= 3 - 1^2 \quad (1.2.25.2)$$

$$= 2 \quad (1.2.25.3)$$

1.2.26 Let  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be such that  $\mathbf{A}^3 = \mathbf{A}$ . Which one of the following statements is ALWAYS correct? (DA 2025)

a)  $\mathbf{A}$  is invertibleb) Determinant of  $\mathbf{A}$  is 0c) The sum of the diagonal elements of  $\mathbf{A}$  is 1d)  $\mathbf{A}$  and  $\mathbf{A}^2$  have the same rank

**Solution:** From (1.1.26.1):

$$\mathbf{A}^3 = \mathbf{A} \quad (1.2.26.1)$$

$$\implies \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^2) \quad (1.2.26.2)$$

To demonstrate that other properties are not universally guaranteed, consider counterexamples satisfying  $\mathbf{A}^3 = \mathbf{A}$ :

- For the zero matrix  $\mathbf{A} = \mathbf{0}$ ,  $|\mathbf{A}| = 0$  and it is not invertible, disproving strict invertibility and confirming determinant can be 0.
- For the identity matrix  $\mathbf{A} = \mathbf{I}_n$ ,  $|\mathbf{A}| = 1 \neq 0$  and the trace is  $n$  (or 2 for  $\mathbf{I}_2$ ), showing that the determinant is not always 0 and the sum of diagonal elements is not universally 1.

1.2.27 Let  $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$  be a set of linearly independent vectors in  $\mathbb{R}^n$ . Let the  $(i, j)$ -th element of matrix  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be given by  $A_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$ ,  $1 \leq i, j \leq n$ . Which one of the following statements is correct? (DA 2025)

a)  $\mathbf{A}$  is invertibleb) 0 is a singular value of  $\mathbf{A}$ c) Determinant of  $\mathbf{A}$  is 0d)  $\mathbf{z}^\top \mathbf{A} \mathbf{z} = 0$  for some non-zero  $\mathbf{z} \in \mathbb{R}^n$ 

**Solution:** From (1.1.27.1), for linearly independent columns of  $\mathbf{X}$ :

$$\mathbf{z}^\top \mathbf{A} \mathbf{z} = \|\mathbf{X} \mathbf{z}\|^2 > 0, \quad \forall \mathbf{z} \neq \mathbf{0} \quad (1.2.27.1)$$

Thus,  $\mathbf{A}$  is strictly positive definite, implying invertibility.

1.2.28 Consider the cumulative distribution function (CDF) of a random variable  $X$ :

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases} \quad (1.2.28.1)$$

The value of  $P(X^2 \leq 0.25)$  is (DA 2025)

a) 0.625

b) 0.25

c) 0.5

d) 0.5625

**Solution:** From (1.1.28.1), with  $c = \sqrt{0.25} = 0.5$ :

$$P(X^2 \leq 0.25) = F_X(0.5) - F_X(-0.5) \quad (1.2.28.2)$$

$$= \frac{1}{4}(0.5 + 1)^2 - \frac{1}{4}(-0.5 + 1)^2 \quad (1.2.28.3)$$

$$= \frac{2.25 - 0.25}{4} \quad (1.2.28.4)$$

$$= 0.5 \quad (1.2.28.5)$$

1.2.29 A random variable  $X$  is said to be distributed as Bernoulli( $\theta$ ), denoted by  $X \sim \text{Bernoulli}(\theta)$ , if  $P(X = 1) = \theta$ ,  $P(X = 0) = 1 - \theta$  for  $0 < \theta < 1$ . Let  $Y = \sum_{i=1}^{300} X_i$ , where  $X_i \sim \text{Bernoulli}(\theta)$ ,  $i = 1, 2, \dots, 300$  be independent and identically distributed random variables with  $\theta = 0.25$ . The value of  $P(60 \leq Y \leq 90)$  after approximation through Central Limit Theorem, is given by (Recall that  $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$ ) (DA 2025)

a)  $\phi(2) - \phi(-2)$ c)  $\phi(3) - \phi(-3)$ b)  $\phi(1) - \phi(-1)$ d)  $\phi(90) - \phi(60)$ 

**Solution:** From (1.1.29.3) and (1.1.29.4), for  $n = 300$  and  $\theta = 0.25$ :

$$\mu = n\theta = 75 \quad (1.2.29.1)$$

$$\sigma = \sqrt{n\theta(1 - \theta)} = 7.5 \quad (1.2.29.2)$$

From (1.1.29.6), with  $a = 60$  and  $b = 90$ :

$$P(60 \leq Y \leq 90) = \phi\left(\frac{90 - 75}{7.5}\right) - \phi\left(\frac{60 - 75}{7.5}\right) \quad (1.2.29.3)$$

$$= \phi(2) - \phi(-2) \quad (1.2.29.4)$$

1.2.30 For  $x \in \mathbb{R}$ , the floor function is denoted by  $f(x) = \lfloor x \rfloor$  and defined as follows

$$\lfloor x \rfloor = k, \quad k \leq x < k + 1 \quad (1.2.30.1)$$

where  $k$  is an integer. Let  $Y = \lfloor X \rfloor$  where  $X$  is an exponentially distributed random variable with mean  $\frac{1}{\ln 10}$ , where  $\ln$  denotes natural logarithm. For any positive integer  $l$ , one can write the probability of the event  $Y = l$  as follows

$$P(Y = l) = q^l (1 - q) \quad (1.2.30.2)$$

The value of  $q$  is

(DA 2025)

a) 0.1

b) 0.01

c) 0.5

d) 0.434

**Solution:** From (1.1.30.1) and (1.1.30.7), with  $\lambda = \ln 10$ :

$$P(Y = l) = q^l (1 - q) \quad (1.2.30.3)$$

$$q = e^{-\lambda} \quad (1.2.30.4)$$

$$= e^{-\ln 10} \quad (1.2.30.5)$$

$$= 0.1 \quad (1.2.30.6)$$

1.2.31 A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event  $A$  represents the set of all outcomes where at least one of the dice shows a 1. Then,  $P(A) =$  (DA 2025)

- a) 0                      b) 1                      c)  $\left(\frac{5}{6}\right)^{100}$                       d)  $1 - \left(\frac{5}{6}\right)^{100}$

**Solution:** From (1.1.31.4), with  $n = 100$  and  $q = 5/6$ :

$$P(A) = 1 - q^n \quad (1.2.31.1)$$

$$= 1 - \left(\frac{5}{6}\right)^{100} \quad (1.2.31.2)$$

1.2.32 Consider the following Python code snippet.

```
A = {"this", "that"}
B = {"that", "other"}
C = {"other", "this"}
while "other" in C:
    if "this" in A:
        A, B, C = A - B, B - C, C - A
    if "that" in B:
        A, B, C = C | A, A | B, B | C
```

When the above program is executed, at the end, which of the following sets contains "this"? (DA 2025)

- a) Only A                      b) Only B                      c) Only C                      d) A, C

1.2.33 Consider the function

$$f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}, \quad x \in [-8, 0] \quad (1.2.33.1)$$

Which of the following statements is/are correct? (DA 2025)

- a) The maximum value of  $f$  is attained at  $x = -5$   
 b) The minimum value of  $f$  is attained at  $x = -2$   
 c) The maximum value of  $f$  is  $\frac{133}{2}$   
 d) The minimum value of the derivative of  $f$  is attained at  $x = -\frac{7}{2}$

**Solution:** From (1.1.32.1), setting  $f'(x) = 0$ :

$$f'(x) = x^2 + 7x + 10 = 0 \quad (1.2.33.2)$$

$$\implies x = -5, x = -2 \quad (1.2.33.3)$$

$$f''(x) = 2x + 7 \quad (1.2.33.4)$$

$$\implies f''(-5) = -3 < 0 \quad (1.2.33.5)$$

$$f''(-2) = 3 > 0 \quad (1.2.33.6)$$

Thus, local maximum occurs at  $x = -5$  and local minimum at  $x = -2$ . The minimum of  $f'(x) = (x + 7/2)^2 - 9/4$  occurs at  $x = -7/2$ .

1.2.34 Let  $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$  be a system of orthonormal vectors in  $\mathbb{R}^{10}$ . Consider the matrix  $\mathbf{A} = \mathbf{x}_1\mathbf{x}_1^\top + \cdots + \mathbf{x}_5\mathbf{x}_5^\top$ . Which of the following statements is/are correct? (DA 2025)

- a) Singular values of  $\mathbf{A}$  are also its eigenvalues
- b) Singular values of  $\mathbf{A}$  are either 0 or 1
- c) Determinant of  $\mathbf{A}$  is 1
- d)  $\mathbf{A}$  is invertible

**Solution:** From (1.1.33.1),  $\mathbf{A}^2 = \mathbf{A}$  and  $\mathbf{A} = \mathbf{A}^\top$ :

$$\lambda \in \{0, 1\} \quad (1.2.34.1)$$

$$\sigma_i = |\lambda_i| = \lambda_i \quad (1.2.34.2)$$

$$\implies \sigma_i \in \{0, 1\} \quad (1.2.34.3)$$

Since  $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = 5 < 10$ ,  $|\mathbf{A}| = 0$ , rendering  $\mathbf{A}$  non-invertible.

1.2.35 Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a twice-differentiable function and suppose its second derivative satisfies  $f''(x) > 0$  for all  $x \in \mathbb{R}$ . Which of the following statements is/are ALWAYS correct? (DA 2025)

- a)  $f$  has a local minima
- b) There does not exist  $x$  and  $y$ ,  $x \neq y$  such that  $f'(x) = f'(y) = 0$
- c)  $f$  has at most one global minimum
- d)  $f$  has at most one local minimum

**Solution:** Since  $f''(x) > 0$  for all  $x \in \mathbb{R}$ , the function  $f(x)$  is strictly convex across the entire real line.

- A strictly convex function has a strictly unique global minimum and at most one local minimum.
- Functions like  $f(x) = e^x$  satisfy  $f''(x) > 0$  everywhere but possess no local minima.
- If  $f'(x) = f'(y) = 0$  for  $x \neq y$ , Rolle's Theorem would imply  $f''(c) = 0$  for some point between them, contradicting the strict condition  $f''(x) > 0$ .

1.2.36 An  $n \times n$  matrix  $\mathbf{A}$  with real entries satisfies the property:  $\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$ , for all  $\mathbf{x} \in \mathbb{R}^n$ , where  $\|\cdot\|$  denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct? (DA 2025)

- a)  $\mathbf{A}$  must be orthogonal
- b)  $\mathbf{A} = \mathbf{I}$  where  $\mathbf{I}$  denotes the identity matrix, is the only solution
- c) The eigenvalues of  $\mathbf{A}$  are either +1 or -1

d)  $\mathbf{A}$  has full rank

**Solution:** From (1.1.34.1):

$$\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2 \quad (1.2.36.1)$$

$$\iff \mathbf{A}^\top \mathbf{A} = \mathbf{I} \quad (1.2.36.2)$$

$\mathbf{A}$  is orthogonal, and hence  $\text{rank}(\mathbf{A}) = n$ .

1.2.37 Consider designing a linear binary classifier  $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$ ,  $\mathbf{x} \in \mathbb{R}^2$  on the following training data:

$$\text{Class-1} : \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}, \quad \text{Class-2} : \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \quad (1.2.37.1)$$

Hard-margin support vector machine (SVM) formulation is solved to obtain  $\mathbf{w}$  and  $b$ . Which of the following options is/are correct? (DA 2025)

a)  $\mathbf{w} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$  and  $b = 1$

b) The number of support vectors is 3

c) The margin is  $\sqrt{2}$

d) Training accuracy is 98%

1.2.38 Consider a coin-toss experiment where the probability of head showing up is  $p$ . In the  $i^{\text{th}}$  coin toss, let  $X_i = 1$  if head appears, and  $X_i = 0$  if tail appears. Consider

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i \quad (1.2.38.1)$$

where  $n$  is the total number of independent coin tosses. Which of the following statements is/are correct? (DA 2025)

a)  $E[\hat{p}] = p$

b)  $E[\hat{p}] = \frac{p}{n}$

c) As  $n$  increases, variance of  $\hat{p}$  decreases

d) Variance of  $\hat{p}$  does not depend on  $n$

**Solution:** From (1.1.35.1) and (1.1.35.2):

$$E[\hat{p}] = p \quad (1.2.38.2)$$

$$\text{var}(\hat{p}) = \frac{p(1-p)}{n} \quad (1.2.38.3)$$

Since  $n$  is in the denominator,  $\text{var}(\hat{p})$  strictly decreases as  $n$  increases.

1.2.39 Let  $G$  be a simple, unweighted, and undirected graph. A subset of the vertices and edges of  $G$  are shown below.

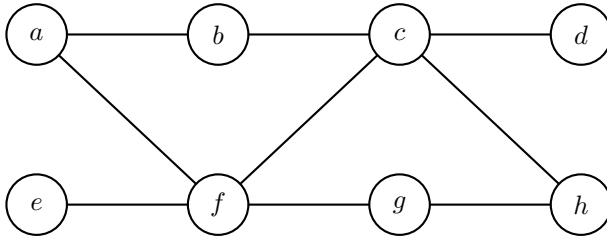


Fig. 1.2.39.1

It is given that  $a - b - c - d$  is a shortest path between  $a$  and  $d$ ;  $e - f - g - h$  is a shortest path between  $e$  and  $h$ ;  $a - f - c - h$  is a shortest path between  $a$  and  $h$ . Which of the following is/are NOT the edges of  $G$ ? (DA 2025)

- a)  $(b, d)$                       b)  $(b, g)$                       c)  $(b, h)$                       d)  $(e, g)$

**Solution:** From (1.1.36.1):

$$d(u, v) \leq d(u, w) + d(w, v) \quad (1.2.39.1)$$

If edge  $(b, d)$  exists:

$$d(a, d) \leq d(a, b) + d(b, d) = 2 < 3 \quad (1.2.39.2)$$

contradicting that  $a - b - c - d$  is a shortest path. If edge  $(e, g)$  exists:

$$d(e, h) \leq d(e, g) + d(g, h) = 2 < 3 \quad (1.2.39.3)$$

contradicting that  $e - f - g - h$  is a shortest path.  $\therefore (b, d)$  and  $(e, g)$  are not edges of  $G$ .

1.2.40 Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be such that  $|f(x) - f(y)| \leq (x - y)^2$  for all  $x, y \in \mathbb{R}$ . Then

$$f(1) - f(0) = \underline{\hspace{2cm}} \quad (1.2.40.1)$$

(DA 2025)

**Solution:** From (1.1.37.1), with  $\alpha = 2 > 1$ :

$$|f(x) - f(y)| \leq |x - y|^2 \quad (1.2.40.2)$$

$$\implies f'(x) = 0 \quad (1.2.40.3)$$

$$\implies f(x) = c \quad (1.2.40.4)$$

$$\implies f(1) - f(0) = 0 \quad (1.2.40.5)$$

1.2.41 Let  $D = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$  be a dataset of  $n$  observations where each  $\mathbf{x}^{(i)} \in \mathbb{R}^{100}$ . It is given that  $\sum_{i=1}^n \mathbf{x}^{(i)} = \mathbf{0}$ . The covariance matrix computed from  $D$  has eigenvalues  $\lambda_i = 100^{2-i}$ ,  $1 \leq i \leq 100$ . Let  $\mathbf{u} \in \mathbb{R}^{100}$  be the direction of maximum variance with  $\mathbf{u}^\top \mathbf{u} = 1$ . The value of

$$\frac{1}{n} \sum_{i=1}^n \left( \mathbf{u}^\top \mathbf{x}^{(i)} \right)^2 = \underline{\hspace{2cm}} \quad (1.2.41.1)$$

**Solution:** From (1.1.38.1), (1.1.38.2), and (1.1.38.3):

$$\max_{\mathbf{u}^\top \mathbf{u}=1} \frac{1}{n} \sum_{i=1}^n \left( \mathbf{u}^\top \mathbf{x}^{(i)} \right)^2 = \max_{\mathbf{u}^\top \mathbf{u}=1} \mathbf{u}^\top \mathbf{C} \mathbf{u} \quad (1.2.41.2)$$

$$= \lambda_{\max} \quad (1.2.41.3)$$

With  $\lambda_i = 100^{2-i}$  for  $1 \leq i \leq 100$ :

$$\lambda_{\max} = \lambda_1 \quad (1.2.41.4)$$

$$= 100^{2-1} \quad (1.2.41.5)$$

$$= 100 \quad (1.2.41.6)$$

1.2.42 A bag contains 5 white balls and 10 black balls. In a random experiment,  $n$  balls are drawn from the bag one at a time with replacement. Let  $S_n$  denote the total number of black balls drawn in the experiment. The expectation of  $S_{100}$  denoted by  $E[S_{100}] = \underline{\hspace{2cm}}$ . (DA 2025)

**Solution:** From (1.1.39.1), with  $p = \frac{10}{15} = \frac{2}{3}$  and  $n = 100$ :

$$E[S_{100}] = \sum_{i=1}^{100} E[I_i] \quad (1.2.42.1)$$

$$= 100p \quad (1.2.42.2)$$

$$= 100 \left( \frac{2}{3} \right) \quad (1.2.42.3)$$

$$= \frac{200}{3} \approx 66.67 \quad (1.2.42.4)$$

1.2.43 Consider the following tables, Loan and Borrower, of a bank.

| Loan     |               |        | Borrower      |          |
|----------|---------------|--------|---------------|----------|
| loan_num | branch_name   | amount | customer_name | loan_num |
| L11      | Banjara Hills | 90000  | Anand         | L11      |
| L14      | Kondapur      | 50000  | Karteek       | L11      |
| L15      | SR Nagar      | 40000  | Karteek       | L14      |
| L22      | SR Nagar      | 25000  | Ankita        | L15      |
| L23      | Balanagar     | 80000  | Gopal         | L19      |
| L25      | Kondapur      | 70000  | Karteek       | L22      |
| L19      | SR Nagar      | 65000  | Karteek       | L23      |
|          |               |        | Sunil         | L23      |
|          |               |        | Sunil         | L25      |

Fig. 1.2.43.1

Query:  $\pi_{\text{branch\_name}, \text{customer\_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch\_name}}(\text{Loan})$  where  $\bowtie$  denotes natural join. The number of tuples returned by the above relational algebra

query is \_\_\_\_\_.

(DA 2025)

**Solution:** From (1.1.40.1):

$$\pi_{\text{branch\_name}}(\text{Loan}) = \{\text{Banjara Hills, Kondapur, SR Nagar, Balanagar}\} \quad (1.2.43.1)$$

No customer in Borrower has loans in all 4 distinct branches:

$$\text{Number of tuples} = 0 \quad (1.2.43.2)$$

1.2.44 Consider the following Python code snippet.

```
def f(a, b):
    if (a == 0):
        return b
    if (a % 2 == 1):
        return 2 * f((a - 1) / 2, b)
    return b + f(a - 1, b)

print(f(15, 10))
```

The value printed by the code snippet is \_\_\_\_\_.

(DA 2025)

1.2.45 Consider the following pseudocode.

```
Create empty stack S
Set x = 0, flag = 0, sum = 0
Push x onto S

while (S is not empty) {
    if (flag equals 0){
        Set x = x + 1
        Push x onto S
    }
    if (x equals 8):
        Set flag = 1
        if (flag equals 1){
            x = Pop(S)
            if (x is odd):
                Pop(S)
                Set sum = sum + x
        }
    }
}

Output sum
```

The value of sum output by a program executing the above pseudocode is \_\_\_\_\_.

(DA 2025)

1.2.46 Consider a directed graph  $G = (V, E)$ , where  $V = \{0, 1, 2, \dots, 100\}$  and  $E = \{(i, j) : 0 < j - i \leq 2 \text{ for all } i, j \in V\}$ . Suppose the adjacency list of each vertex is

in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is \_\_\_\_\_. (DA 2025)

**Solution:** From (1.1.41.2), for  $0 \leq k \leq 50$ :

$$D(k) = 2k \quad (1.2.46.1)$$

$$\implies 2k = 50 \quad (1.2.46.2)$$

$$\implies k_{50} = 25 \quad (1.2.46.3)$$

From (1.1.41.3), the remaining discoveries evaluate to:

$$n - k_{50} = 100 - 25 \quad (1.2.46.4)$$

$$= 75 \quad (1.2.46.5)$$